The Landscape of Keyboard Layouts

Local-Optima Networks and Quality Diversity

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Working draft skeleton · no results yet · bracketed TODOs are placeholders
Focus paper 1. Topics T3, T4. Status: tooling complete (`explore`, `atlas`, `qd`, `nsga2`); runs pending.

Abstract

We characterize the search space of keyboard layouts ($30! \approx 2.65 \times 10^{32}$ placements) with a random-walk correlation analysis, a local-optima network sampled by breakout local search, and a MAP-Elites map over ergonomic behaviours. [TODO: fitness-distance correlation value; number of funnels; fraction of local optima within 1 % of best; Pareto front shape.] Popular community layouts [TODO: lie on / behind] the front. We release the atlases and a viewer.

1. Research questions

RQ1 Big-valley or multi-funnel? (FDC, funnel decomposition.) RQ2 How many distinct near-optimal layouts exist and how different are they (Hamming, finger distance, swap distance)? RQ3 What do you pay to get more rolls or less index load (QD, Pareto)? RQ4 Where do the ~4,100 community layouts sit on the map?

2. Hypotheses

H1 positive FDC (big valley). H2 a few funnels differing in vowel placement. H3 the number of near-optimal optima $\gg$ the number of used reference layouts. H4 the roll/SFB front is short and convex.

3. Method

`gecko landscape` (random distribution, correlation length), `gecko explore` (10 min $\times$ 10 seeds per setting), `gecko atlas` (families, consensus, funnels), `gecko qd --axes roll_ratio,index_load --bins 16`, `gecko optimize --algo nsga2`. Canonicalize under left/right mirroring only when provably symmetric. Community layouts from `data/community/cmini.json`, scored on the same tables.

4. Existing evidence

Space statistics and distance definitions (`research/studies/search-space-visualization.md`); exploration and QD specs (`context/algorithms/explore.md`, `map-elites.md`).

5. Figures

F1 local-optima network · F2 FDC · F3 score density vs optima · F4 autocorrelation · F5 consensus keyboard and contested slots · F6 MAP-Elites heatmap with references and community density · F7 Pareto fronts.

6. Threats

Sampling bias of BLS (kicks define the network edges); results are for one profile (see paper 2); the descriptor choice shapes the QD map.